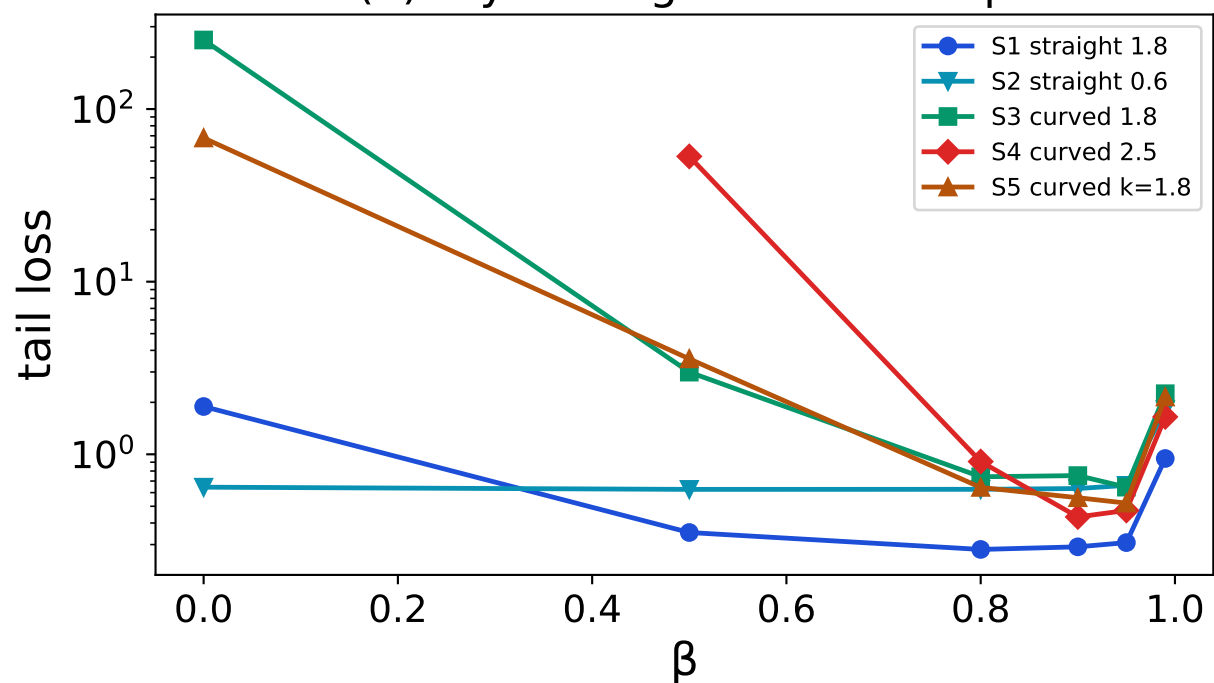
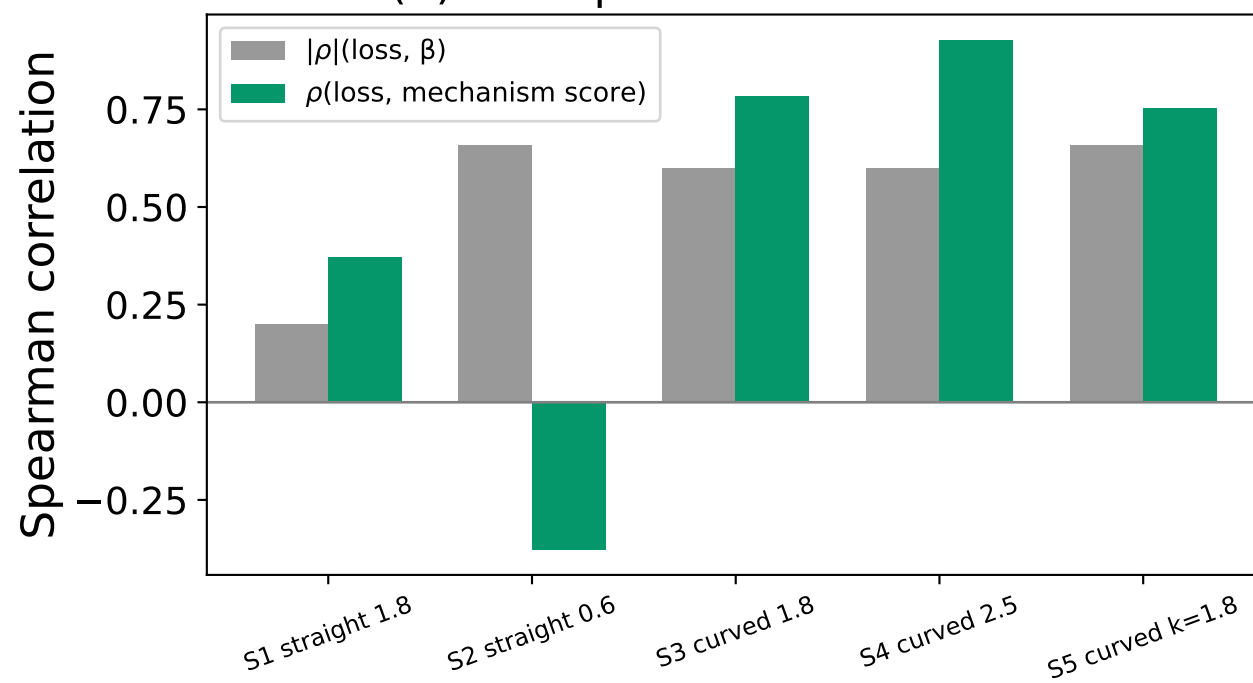
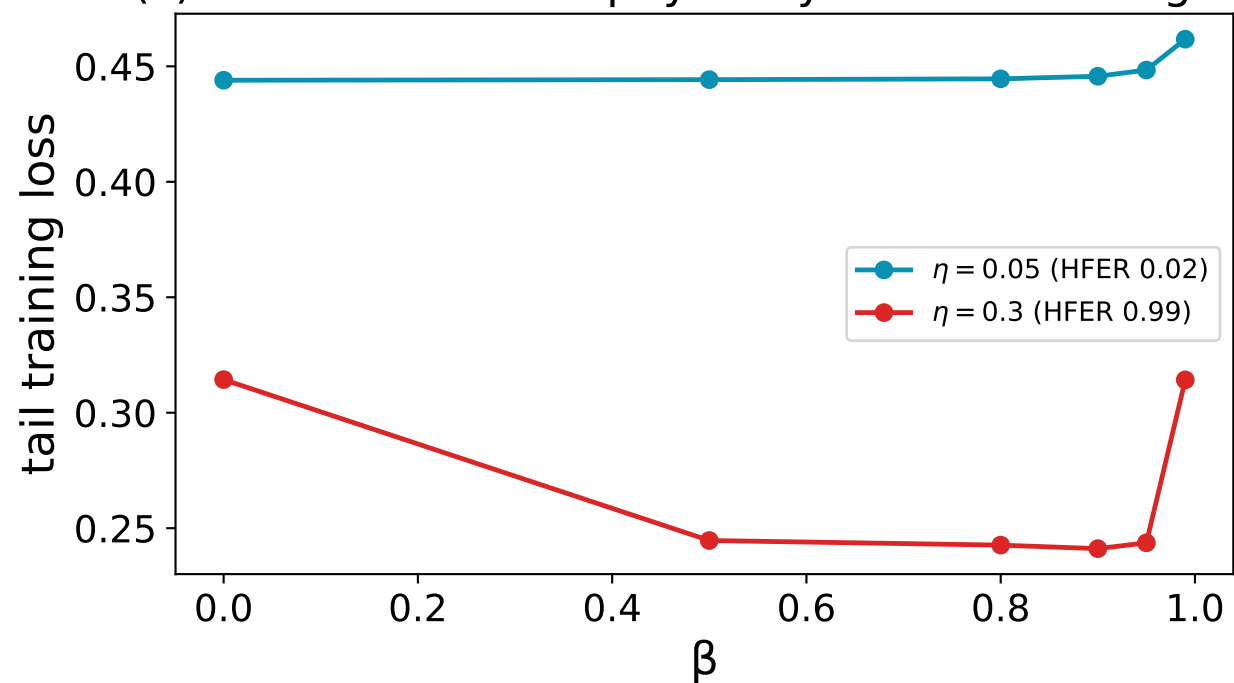


(a) toy settings: tail loss vs β 

(b) what predicts the loss rank



(c) MLP: momentum pays only in the HFER regime



(d) Muon closed loop: the pre/post gap follows the regime

